From Molecular Trajectories to Composable Paths

Alanine dipeptide and the chignolin variant CLN025

Joseph Campagna, PSM October 2026

Abstract. Can molecular geometry make the composition of dynamical behavior visible? We construct an interactive atlas from 750,000 public molecular dynamics (MD) frames of alanine dipeptide and 63,999 frames of the ten-residue chignolin variant CLN025. Molecular envelopes form landscapes over torsional or geometric observables. Two distinct categorical nerves can be superimposed: an overlap face-poset nerve and a free path-category nerve derived from observed basin passages. Translucent pathway skins now deform through an atom-attached displacement field, rather than merely changing illumination. This construction is a visual experiment and a precise vocabulary for discussing composition. It is not a claim of recovered configuration-space topology, a validated Markov model, or a completed theory of compositional protein dynamics. The prototype motivates an experimentally testable route from trajectory composition toward composition of open dynamical subsystems.

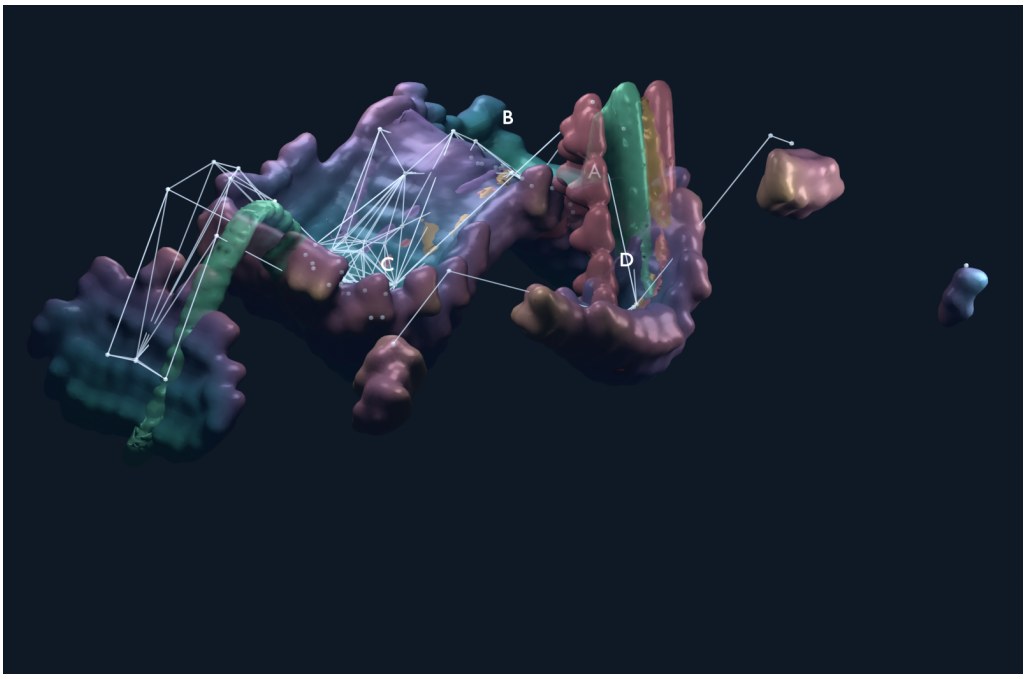


Figure 1: **A molecular landscape with a superimposed nerve.** Blender 4.0/Eevee rendering of the viewer's meshes and deformation targets. Terrain color follows empirical relative free energy (mint: populated; violet/amber: less populated). Pathway skins use source-basin colors. Silver nodes and edges show the overlap face-poset nerve. Labels A–D identify density modes. Horizontal coordinates encode ϕ, ψ; vertical placement encodes $1.3\,F/k_BT$ plus molecular relief. Scene distances are visual units.

Explore the artifact. Interactive molecular atlas and public source repository. The viewer supports deformation amplitude, transparency, route filtering, scrubbing, and independent nerve overlays. All figures are generated from the numerical geometry; they are not AI-generated illustrations.

Molecular systems and cartoon conventions

The two cases share a categorical construction, but differ in molecular resolution and chart coordinates. Alanine dipeptide is represented by ten heavy atoms. CLN025 is represented by ten Cα sites, one per residue: the equal site counts do not imply equal molecular complexity.

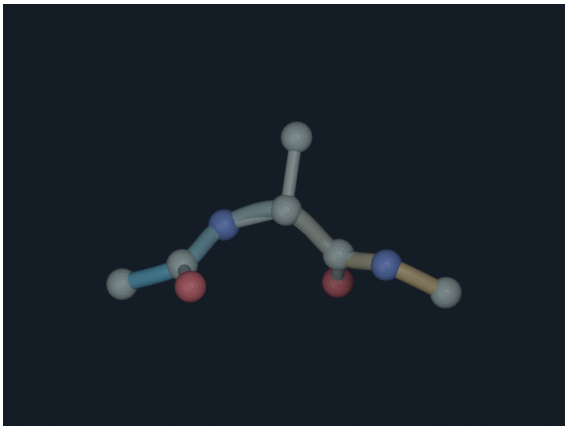


Figure 2: **Alanine dipeptide: a backbone cartoon with atomic context.** The blue-to-gold tube traces the capped backbone from one end to the other; heavy atoms and bonds retain chemical context (oxygen red, nitrogen blue). Coordinates come from the recorded representative of overlap object U_0. A helix/sheet ribbon would not be meaningful for this short benchmark.



Figure 3: **Chignolin CLN025: compact and extended Cα cartoons.** Sequence YYDPETGTWY. Actual frames nearest the 5th and 95th percentiles of Cα radius of gyration are shown. Ribbons interpolate the recorded Cα sites; blue-to-gold indicates N-to-C order. These are geometric backbone cartoons, not DSSP secondary-structure assignments or assertions of native and unfolded states. Each panel has its own fitted camera scale.

The cartoons and landscape figures were rendered in Blender from exported MD coordinates. Smooth ribbons and tubes are illustrative interpolation between sites. The CLN025 mutation and capping conventions are inherited from the deposited simulation topology; the example is not mislabeled as wild-type chignolin. The viewer provides corresponding backbone-trace insets, recorded representatives, and animated route conformations.

1. Two nerves, two meanings of composition

For alanine, let $q_t = (\phi_t, \psi_t) \in \mathbb{T}^2$ describe the backbone angles, and let $X_t \in \mathbb{R}^{10\times 3}$ contain the aligned heavy-atom coordinates. Alanine dipeptide is a capped amino-acid benchmark, not a folded protein. Its small conformational chart makes the distinctions below inspectable.

Overlap category. Take 18 balls of radius 73° in a cut angular chart. A finite sample W (every tenth MD frame) witnesses common membership:

$$K_W = \{\sigma \neq \varnothing : \exists q \in W,\ q \in \bigcap_{i\in\sigma} U_i\}.$$

We retain its 2-skeleton $K_W^{(2)}$ and form the face-poset category $\mathcal{C}_\cap = P(K_W^{(2)})$. Objects are simplices; arrows are inclusions of index sets, not reverse inclusions of intersection regions. Its nerve has 91 objects, 282 strict inclusions, and 204 nondegenerate 2-simplices. The realization is a barycentric subdivision:

$$|\mathrm{N}P(K_W^{(2)})| \cong |K_W^{(2)}|.$$

This categorical statement does not establish equivalence with the protein's configuration space. Good-cover hypotheses and sampling guarantees are separate issues [2, 3]. The overlap chart does not glue its periodic seam.

Dynamic path category. Let G have four basin objects and ten directed generators, one for each observed route type (CLN025 has twelve generators). Set $\mathcal{C}_{\mathrm{path}} = \mathrm{Free}(G)$. Morphisms are finite composable words; identities are empty words. For any category,

$$\mathrm{N}(\mathcal{C})_n = \mathrm{Fun}([n], \mathcal{C}) = \{x_0 \xrightarrow{f_1} \cdots \xrightarrow{f_n} x_n\}.$$

Inner face maps compose adjacent arrows; degeneracies insert identities. For a 2-simplex (f, g), $d_0(f, g) = g$, $d_1(f, g) = g \circ f$, and $d_2(f, g) = f$. Thus the long edge is a composite, not a missing observation to be filled in.



Figure 4: **The two displayed simplicial meanings.** Left: composition of inclusions. Right: formal composition of route generators. An independently observed $A \to C$ generator h_{AC} remains a different arrow. The displayed 25 alanine pairs (36 for CLN025) are only a finite view of the infinite path-category nerve. Return paths are not identities.

The two categories coexist as overlays. No functor identifying them has been constructed. Likewise, a composable pair need not occur consecutively in the recorded MD trajectory.

2. Empirical dynamics before categorical interpretation

The MDshare benchmark contains three independent 250 ns trajectories at 300 K, with 1 ps frame spacing, generated using ACEMD, AMBER ff99SB-ILDN, and TIP3P water [1]. We use the supplied angles and aligned heavy-atom positions. All 750,000 frames contribute to occupancy estimates. Equal-area 12° angular bins are smoothed periodically with a Gaussian of width 0.85 bins:

$$\widehat{F}(q)/k_BT = -\log \frac{\widehat{p}(q)}{\max_{q'} \widehat{p}(q')}.$$

Pooling is unweighted. This is an empirical occupancy landscape; equilibrium convergence and estimator uncertainty have not been established.

Density modes are found separately on a 60×60 grid, smoothed at 1.4 bins. Disjoint 25° cores around four peaks define passage labels. The label ℓ_t retains the last core visited between core entries. Fifteen initial frames, before any core visit, are excluded from labeled counts.

Basin	ϕ	ψ	Last-core occupancy
A	−75°	153°	39.57%
B	−141°	159°	25.62%
C	−75°	−15°	31.82%
D	51°	33°	2.99%

These are density basins, not demonstrated metastable macrostates. In particular, A and B interconvert rapidly. Across independent runs, 73,145 core-to-core passages produce ten directed route types. Representative counts are A→B: 34,316; B→A: 34,323; A→C: 1,176; B→D: 1; and D→C: 2. Sparse directions remain visible and explicitly flagged.

Lagged probabilities. For lag $\tau \in \{10, 50, 100\}$ ps, count sliding pairs within each run:

$$C_{ij}(\tau) = \sum_r \sum_{t \in I_r(\tau)} \mathbf{1}\{\ell_t^{(r)} = i, \ell_{t+\tau}^{(r)} = j\}, \qquad \widehat{P}_{ij}(\tau) = \frac{C_{ij}(\tau)}{\sum_k C_{ik}(\tau)}.$$

Here $I_r(\tau)$ includes only valid labeled pairs in run r. Self transitions are included; pairs never cross run boundaries. Lagged probabilities may include indirect routes. Sliding observations are correlated, so row counts are not independent sample sizes. Gross passage frequency $n_{ij}/750\,\text{ns}$ is distinct from net current.

What has been checked. Independent counts obtained by removing non-core observations and counting changes reproduce the passage counts. Exported atomic frames match the raw coordinates after centering. The first 100 raw coordinate frames reproduce the supplied torsions to about 2.4×10^{-7} radians. Probability rows normalize; periodic-distance invariance, endpoint recovery and within-run lag accounting pass automated checks.

What remains unvalidated. There is no Chapman–Kolmogorov test, implied-timescale plateau, block-bootstrap uncertainty estimate, or detailed-balance assessment. The atlas should not be used to infer new rates, free-energy barriers or mechanistic causality.

3. Chignolin CLN025: extending the atlas

The second case uses the CC BY 4.0 *Chignolin Simulations* deposit [7]. Its full dataset comprises 3,744 adaptive-sampling trajectories of nominal duration 50 ns (187.2 μs aggregate), generated with ACEMD, CHARMM22*, TIP3P water and a 350 K Langevin thermostat. For a bounded, reproducible prototype, we select the first 128 non-hidden XTC files in archive order. These contain 63,999 frames: 127 runs have 500 frames and one has 499. The selected nominal aggregate duration is 6.4 μs; it is neither a continuous trajectory nor an unbiased ensemble.

Coordinates and chart. Ten Cα positions are converted from Å to nm, centered, and rigidly aligned to the first recorded frame by a proper Kabsch rotation. Neighboring Cα distances are checked to lie in $(0.3, 0.5)$ nm. Define geometric, equal-site-weight observables

$$R_g(X) = \sqrt{\frac{1}{10}\sum_{a=1}^{10} \|X_a - \bar{X}\|^2}, \qquad d_{\mathrm{ee}}(X) = \|X_{10} - X_1\|, \qquad q(X) = (R_g, d_{\mathrm{ee}}) \in \mathbb{R}^2_{\geq 0}.$$

These observables do not require alignment, although consistent alignment is needed to place molecular shapes in the scene. The observed coordinate range, padded by 0.01 nm, is mapped linearly to a 30×30 visual chart. A 30×30 histogram is Gaussian-smoothed at 0.85 bins with reflecting boundaries. The displayed score $E_{\mathrm{occ}}(q) = -\log[\widehat{p}(q)/\widehat{p}_{\max}]$ is descriptive negative log occupancy, *not* an equilibrium free energy.

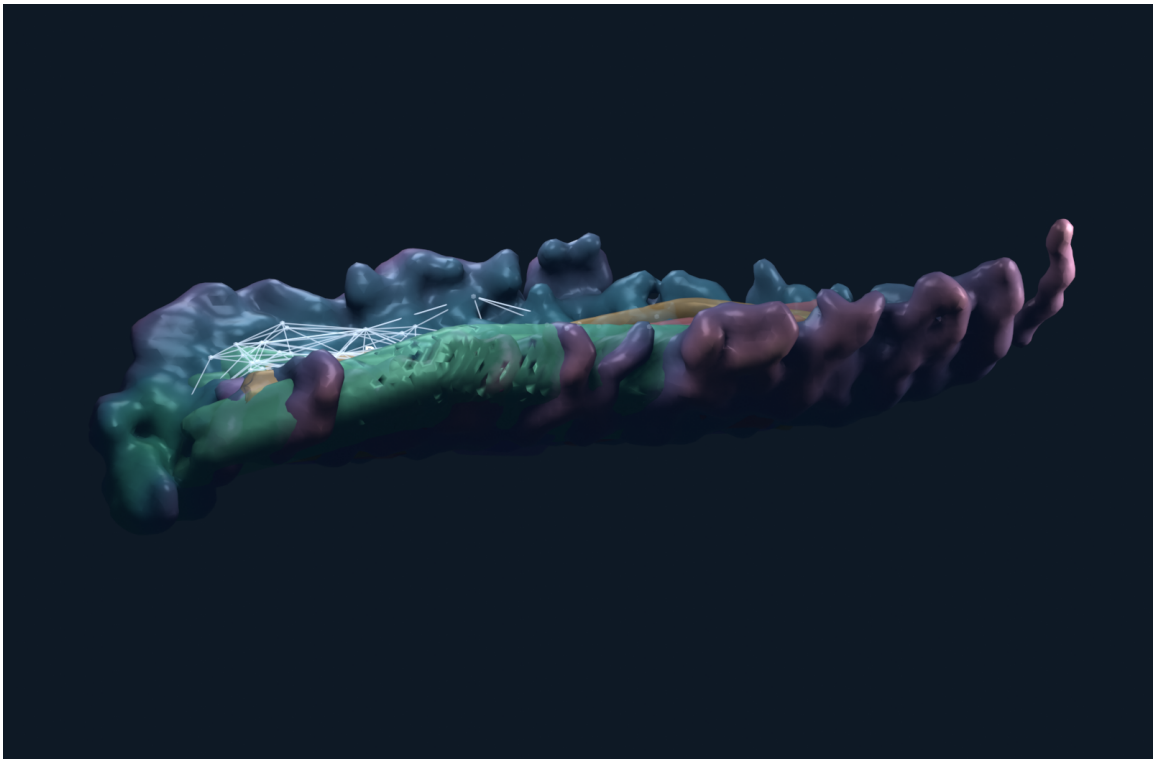


Figure 5: **The CLN025 molecular landscape and overlap nerve.** Ten-site backbone geometries generate the terrain and deforming pathway envelopes. Silver lines show inclusions in the overlap face poset. Horizontal axes encode R_g and d_{ee}; height combines $1.3E_{\mathrm{occ}}$ with molecular relief. This is a visualization of the selected adaptive sample, not a folding free-energy surface.

Chignolin transitions and comparison

Four k-means centers are fitted to every fifth observation in standardized (R_g, d_{ee}) coordinates, using seed 42 and 50 iterations. Centers are ordered by increasing R_g and labeled A–D. A frame lies in a core when its nearest-center distance is at most 0.42. The last visited core labels intervening observations. The first 129 pre-core observations are excluded; label memory resets at each run boundary. There are 6,979 observed core-to-core passages and twelve directed route types. These clusters are descriptive geometric states, not validated metastable folding macrostates.

Quantity	Alanine dipeptide	CLN025 subset
Recorded frames	750,000	63,999
Independent runs	3	128
Temperature	300 K	350 K
Displayed molecular sites	10 heavy atoms	10 Cα sites
Core-to-core passages	73,145	6,979
Observed route types	10	12
Overlap nerve objects	91	97
Strict inclusion arrows	282	342
Overlap 2-simplices	204	264
Composable generator pairs	25	36

Counts compare software constructions, not rates or equivalent dynamical complexity. Sampling, temperature, molecular resolution and state definitions differ.

A finite overlap nerve. Fourteen farthest-point centers in the rescaled chart define radius-5.2 Euclidean balls. Every fifth observation witnesses common membership. As for alanine, the face-poset nerve is formed from the witnessed 2-skeleton. This chart metric is a declared visualization choice; it is not a physical distance between protein conformations.

Pathways and skins. Each route uses an actual within-run passage nearest its route's median duration in frames. At most 80 recorded conformations are retained, and 96 linearly interpolated samples define its envelope. Cα coordinates are magnified by 2.3 visual units/nm; Gaussians have width 0.43, grid spacing 0.30, threshold 0.65, and 0.45-voxel smoothing. The same bounded material warp as alanine supplies twelve position/normal targets.



Figure 6: **CLN025 A→C skin at visual phases** $0, 1/3, 2/3$**.** Only this route is shown, under fixed camera and lighting. Surface motion comes from recorded Cα geometry but is not physical time evolution of a material molecular surface.

4. From swept molecular skins to deforming geometry

For a representative passage, let $u \in [0,1]$ parameterize its recorded time. Coordinates are centered and scaled by 3.5 visual units per nm, and translated to $(\phi/12,\ 1.3\widehat{F}/k_BT,\ \psi/12)$. Interpolated atomic Gaussians of width 0.36 visual units form a swept envelope:

$$\rho_u(x) = \sum_a \exp\left[-\frac{\|x - Y_a(u)\|^2}{2(0.36)^2}\right], \qquad S = \{x : \max_u \rho_u(x) = 0.7\}.$$

The implementation uses at least 96 temporal samples, a 0.22-unit grid, and 0.45-voxel smoothing before isosurface extraction. Skins share the terrain's height and molecular scale; they are translucent intersecting meshes, not a remeshed Boolean union. Cut-chart crossings are not bridged.

Actual vertex motion. Each skin vertex v receives an approximate nearest-atom phase u_v and normalized atom weights $w_{va} \propto \exp[-\|v - Y_a(u_v)\|^2/(2(0.7)^2)]$. We introduce a bounded material time warp:

$$\tau_t(u) = u + 0.08\sin(2\pi(t-u))\sin(\pi u), \qquad D_v(t) = \sum_a w_{va}\left[Y_a(\tau_t(u_v)) - Y_a(u_v)\right].$$

It fixes the passage endpoints and remains within $[0,1]$. After two adjacency-averaging steps L, the rendered vertex is

$$v(t) = v + B_{1.5}\Big(L^2 D_v(t)\Big), \qquad B_c(z) = c\tanh(\|z\|/c)\frac{z}{\|z\|}, \quad B_c(0) = 0.$$

Endpoint-attached displacements are reset to zero. Twelve cyclic keyframes store both positions and recomputed normals; the browser interpolates between neighboring targets. A deformation-amplitude control scales the displacement down to zero for direct comparison with the fixed-envelope version.

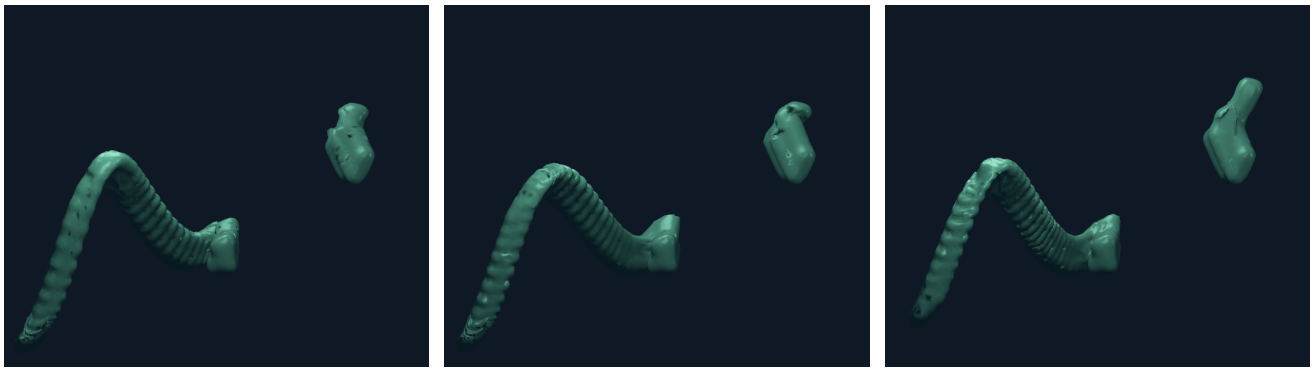


Figure 7: **Geometry changes, not only illumination.** Blender snapshots of the A→C skin at visual phases $t = 0, 1/3, 2/3$, with the same camera and lighting. Other route skins, terrain and the nerve are hidden. These phases are material deformation phases, not three complete instantaneous MD configurations.

This is an artistic, MD-attached deformation field. Interpolated Cartesian coordinates need not preserve bond lengths; self-overlaps can have ambiguous phase assignments. We do not assert injectivity, volume conservation, topology preservation, physical forces, or a dynamical semigroup for this animation.

5. Limitations and obstacles encountered

This section records the substantive scientific and practical obstacles encountered during construction, the response taken, and the remaining limits on interpretation. Successful rendering or passing numerical checks does not resolve these scientific limitations.

Dataset scale and selection. The CLN025 archive is approximately 1.41 GB and contains thousands of runs. We streamed the first 128 trajectory members to keep the example reproducible on ordinary hardware. Archive order is deterministic but not random or statistically representative. Only per-member hashes were verified; the full archive checksum was not independently verified. One selected run contains 499 frames rather than 500, so all code uses actual lengths and resets at run boundaries. Expanding or randomizing the selection may change every reported cluster, passage count and occupancy.

Timing conflict. The first selected XTC encodes a 100,000 ps interval and a final time of 50,000,000 ps, whereas the deposit describes 50 ns production runs. This is a factor-1,000 discrepancy relative to a nominal 100 ps frame interval. We preserve raw per-file times, analyze passages in frame indices, and label 6.4 μs only as the nominal aggregate from deposit metadata. We do not silently repair the files, infer rates from their timestamps, or claim that the discrepancy has been resolved.

Adaptive sampling and comparability. CLN025 trajectories were adaptively initialized, and no equilibrium weighting is available in this prototype. Pooled occupancy is therefore not a justified Boltzmann probability. Alanine also lacks a convergence and uncertainty study. The systems have different temperatures, force fields, atom selections and sampling procedures. Their plotted heights and transition counts must not be compared as quantitative thermodynamic or kinetic measurements.

Chemical identity and cartoons. The available deposit is CLN025 (YYDPETGTWY), not original chignolin. Its topology includes non-protein entries that trigger element/mass warnings in the trajectory parser. We select Cα atoms by name, verify the ten-residue sequence, and use no guessed masses or solvent atoms in the geometric features. Alanine cannot support a conventional protein secondary-structure cartoon; its backbone trace includes atomic context. CLN025 ribbons interpolate Cα sites without assigning helices, sheets, native state or hydrogen bonds.

Projection and state definitions. (R_g, d_{ee}) can merge folded, misfolded and otherwise distinct structures. Alignment to one reference is a rendering gauge, not a dynamical reduction. Core radii, smoothing widths, cover size and chart scaling are selected parameters, without a robustness study. The two-dimensional state labels do not establish metastability, sufficient statistics, memorylessness, or a reaction coordinate.

Limitations: topology, deformation and implementation

Categorical claims. The identity between the realization of a face-poset nerve and the barycentric subdivision of its simplicial complex is exact. Recovering the homotopy type of molecular configuration space would require additional cover and sampling hypotheses, which are not established. Witnessing intersections in a finite projection may miss or merge features. The alanine seam is unglued. The overlap complex is truncated, while the free path category has an infinite nerve; the viewer shows only selected finite parts. No comparison functor between the two categories, proof-assistant export, or compositional semantics for molecular dynamics has been implemented.

Statistical and dynamical validation. Some alanine directions have one or two passages. Temporal correlations, core recrossings and arbitrary clustering affect counts in both systems. The chignolin display provides passage counts, not a validated Markov model. No implied-timescale convergence, Chapman–Kolmogorov test, bootstrap uncertainty, detailed-balance analysis or experimental validation is supplied. Matching categorical endpoints permits a formal composition even when its constituent passages were never recorded consecutively.

Deformation and geometry. Fixed swept skins initially communicated motion only through a highlight. The revision adds actual position/normal morph targets, but this remains a visualization construction. Local attachment can be ambiguous at self-overlaps. Cartesian interpolation can distort molecular geometry; adjacency smoothing and a 1.5-unit displacement bound reduce extreme excursions without guaranteeing injectivity, collision freedom, conserved volume or preserved topology. Envelopes intersect the terrain as separate transparent meshes rather than forming a Boolean union. Transparency ordering can hide surfaces. Disconnected islands are possible consequences of sparse sampling and thresholded densities, not newly discovered physical states.

Rendering and software constraints. Initial figure framing clipped isolated pathway skins; cameras were refitted to all deformation targets before final rendering. Twelve targets and bounded representative samples keep assets manageable, at the cost of temporal and spatial detail. GPU limits and transparency behavior vary between browsers. Blender figures use 4.0/Eevee; other releases may change material APIs or output. A trial MDTraj installation required a source build and subsequently lacked an import dependency. The pipeline instead uses the functioning MDAnalysis parser, with a documented Python environment. The static viewer requires HTTP serving, not direct `file://` loading.

Publication and reproducibility scope. The repository originally mixed this atlas with a separate polyethylene simulation project. Documentation now separates them; generated raw simulations are not required to run the atlas. Previously tracked inputs, build binaries and smoke-run artifacts were removed from the current source index while preserving local files and Git history; older commits still contain those artifacts. Large upstream MD inputs are downloaded on demand and have separate attribution. The hosted Sites viewer has its own access policy; public GitHub source does not automatically make that hosted instance public. A local server or the public repository's documented viewer deployment is the reproducible entry point. This paper is exploratory and not peer reviewed; implementation tests establish consistency, not scientific validity.

6. Future directions

The current construction makes *path composition* inspectable. Composing complete dynamical subsystems is a stronger objective. For an autonomous flow, one expects $\Phi_{t+s} = \Phi_t \circ \Phi_s$; a Markov approximation would analogously require $P(t+s) \approx P(t)P(s)$ under a stated convention. Neither property follows from drawing a nerve.

One proposed next step is to define structural interfaces (for example, selected contact or torsional observables), attach validated dynamics to modules, and specify how compatible interfaces compose. Open reaction networks and open Markov processes provide established categorical examples: input/output boundaries allow systems to compose, and functorial semantics can relate composite models to their behavior [4, 5]. Applying such ideas to structural bioinformatics would require new choices and validation, not just a change in visualization.

The research questions are concrete:

1. Can basin definitions and transition estimates remain stable across trajectory blocks, temperatures, and resolutions?
2. Can experimentally meaningful interfaces support compositional models with quantified approximation error?
3. Can a semantics functor $\mathcal{S} : \mathcal{D} \to \mathcal{B}$ satisfy $\mathcal{S}(g \circ f) = \mathcal{S}(g) \circ \mathcal{S}(f)$, where $\mathcal{D}$ is a category of specified open models and $\mathcal{B}$ their behaviors?
4. For CLN025, can native-contact, hydrogen-bond and backbone-torsion features distinguish compact misfolds from the native hairpin? Compare the current two-observable chart with time-lagged coordinates before interpreting any cluster as a folding state.
5. Can the timestamp discrepancy be resolved with the depositors and the complete adaptive dataset be analyzed with justified statistical weights? Then test implied timescales, Chapman–Kolmogorov consistency and held-out transition predictions, with uncertainty from independent runs.
6. Can an explicitly defined projection between atomistic and Cα state categories preserve specified compositions? Such a comparison functor, including its approximation error, would connect the two visualization scales.

Here $\mathcal{D}$, $\mathcal{B}$, and $\mathcal{S}$ are research targets, not artifacts implemented by this prototype. Homotopy.io offers a useful reference for higher-categorical diagramming [6]; the current viewer is a custom Three.js implementation, without a homotopy.io export or proof certificate.

Reproducibility. The companion source includes data-download and analysis scripts, MD-driven morph targets, the LaTeX source, a Blender scene with shape keys, and the figure-generation script. The browser and Blender use the same geometry. The underlying source-file SHA-256 hashes are recorded in `dist/data.json`. Regenerate in this order: `download.py`, `prepare.py`, `dynamics.py`, `build_skins.py`, `build_deformation.py`. For CLN025, run `download_chignolin.py` then `build_chignolin.py`. The repository provides a static viewer, runnable tests and both figure pipelines. The figures use Blender 4.0/Eevee; the document is compiled from LaTeX.

Exploratory proposition. Categorical language may help separate state, transition, composition, and interface in molecular modeling. This atlas offers a visible, inspectable starting point for that discussion, rather than evidence that categorification has already improved a predictive model.

7. Availability, attribution and references

Software and paper. https://github.com/joecampdb/pronerv_MTNerv. The molecular atlas lives in `alanine-nerve/`; its historical directory name is retained for compatibility. The repository also retains an independent polyethylene screening workflow, documented separately. The compiled paper is `paper/molecular-atlas.pdf` at repository root.

Data and transformations. The CLN025 data are attributed to Adrià Pérez Culubret and Gianni De Fabritiis under CC BY 4.0 [7]. The subset is streamed in archive order; individual PDB/XTC SHA-256 hashes and exact member names are retained. Transformations include Cα selection, centering, rigid alignment, geometric features, clustering, density estimation, subsampling, and visual surface extraction. The original archive is approximately 1.41 GB; only the selected members are read. The deposit's archive MD5 is recorded as metadata, not asserted to have been verified against a complete local download.

Interpretation of the two examples. Alanine is a torsional benchmark; CLN025 is a ten-residue mini-protein variant. The extension demonstrates that the software can represent a more complicated conformational dataset, not that the same two-dimensional representation resolves its folding kinetics. No quantitative performance gain from categorification has been established.

Authorship and preparation. Joseph Campagna, PSM. This exploratory software and manuscript were prepared with AI-assisted coding and drafting. Numerical outputs and figure sources are included for inspection; the document has not undergone peer review.

References

[1] MDshare. *Alanine dipeptide dataset and simulation metadata.* https://markovmodel.github.io/mdshare/ALA2/.

[2] J. P. May. *Simplicial sets, nerves of categories, Kan complexes, etc.* https://math.uchicago.edu/~may/REU2018/sset.pdf.

[3] U. Bauer, M. Kerber, F. Roll, A. Rolle. *A Unified View on the Functorial Nerve Theorem and its Variations.* https://arxiv.org/abs/2203.03571.

[4] J. C. Baez, B. S. Pollard. *A Compositional Framework for Reaction Networks* (2017). https://arxiv.org/abs/1704.02051.

[5] B. S. Pollard. *Open Markov Processes and Reaction Networks* (2017). https://arxiv.org/abs/1709.09743.

[6] N. Corbyn et al. *homotopy.io: a proof assistant for finitely-presented globular n-categories* (2024). https://arxiv.org/abs/2402.13179.

[7] A. Pérez Culubret and G. De Fabritiis. *Chignolin Simulations* (2021), Figshare, CC BY 4.0. https://doi.org/10.6084/m9.figshare.13858898.